

Manual Change History (4021375) QSK23 Troubleshooting and Repair Manual

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Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-022-001_16 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_16.html)
Procedure Title	Service Tools
Update Reason	

Date	19-Jun-2003
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-014-008 (/qs3/pubsys2/xml/en/procedures/89/89-014-008.html)
Procedure Title	Engine Testing (In Chassis)
Update Reason	
Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-005 (/qs3/pubsys2/xml/en/procedures/89/89-001-005.html)
Procedure Title	Bearings, Connecting Rod
Update Reason	
Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-006 (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)
Procedure Title	Bearings, Main
Update Reason	
Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-007 (/qs3/pubsys2/xml/en/procedures/89/89-001-007.html)
Procedure Title	Bearings, Thrust
Update Reason	
Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-010 (/qs3/pubsys2/xml/en/procedures/89/89-001-010.html)
Procedure Title	Camshaft Bushings
Update Reason	

Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-011 (/qs3/pubsys2/xml/en/procedures/89/89-001-011.html)
Procedure Title	Camshaft Cover Plate
Update Reason	

Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-013 (/qs3/pubsys2/xml/en/procedures/89/89-001-013.html)
Procedure Title	Camshaft Gear (Camshaft Removed)
Update Reason	

Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-020 (/qs3/pubsys2/xml/en/procedures/89/89-001-020.html)
Procedure Title	Crankshaft Gear, Rear (Crankshaft Installed)
Update Reason	

Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-023 (/qs3/pubsys2/xml/en/procedures/89/89-001-023.html)
Procedure Title	Crankshaft Seal, Front
Update Reason	

Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-024 (/qs3/pubsys2/xml/en/procedures/89/89-001-024.html)
Procedure Title	Crankshaft Seal, Rear
Update Reason	

Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-026 (/qs3/pubsys2/xml/en/procedures/89/89-001-026.html)
Procedure Title	Cylinder Block
Update Reason	
Date	20-Jun-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-036 (/qs3/pubsys2/xml/en/procedures/89/89-001-036.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	
Date	20-Jun-2003
Section	Section 11 - Exhaust System - Group 11
Procedure Number	89-022-001_11 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_11.html)
Procedure Title	Service Tools
Update Reason	
Date	20-Jun-2003
Section	Section 11 - Exhaust System - Group 11
Procedure Number	89-011-001 (/qs3/pubsys2/xml/en/procedures/89/89-011-001.html)
Procedure Title	Exhaust Connection Pipe
Update Reason	
Date	20-Jun-2003
Section	Section 11 - Exhaust System - Group 11
Procedure Number	89-011-007 (/qs3/pubsys2/xml/en/procedures/89/89-011-007.html)
Procedure Title	Exhaust Manifold, Dry
Update Reason	

Date	20-Jun-2003
Section	Section 11 - Exhaust System - Group 11
Procedure Number	89-011-009 (/qs3/pubsys2/xml/en/procedures/89/89-011-009.html)
Procedure Title	Exhaust Restriction
Update Reason	
Date	20-Jun-2003
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-022-001_13 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_13.html)
Procedure Title	Service Tools
Update Reason	
Date	20-Jun-2003
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-020-tr (/qs3/pubsys2/xml/en/procedures/89/89-013-020-tr.html)
Procedure Title	Starting Motor
Update Reason	
Date	20-Jun-2003
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-014-004 (/qs3/pubsys2/xml/en/procedures/89/89-014-004.html)
Procedure Title	Engine Run-in (Without Dynamometer)
Update Reason	
Date	08-Jul-2003
Section	Section E - Engine Identification
Procedure Number	89-100-002 (/qs3/pubsys2/xml/en/procedures/89/89-100-002.html)
Procedure Title	Engine Diagrams
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-001-999 (/qs3/pubsys2/xml/en/procedures/89/89-001-999.html)
Procedure Title	Cylinder Block - Overview
Update Reason	
Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-002-999 (/qs3/pubsys2/xml/en/procedures/89/89-002-999.html)
Procedure Title	Cylinder Head - Overview
Update Reason	
Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-003-999 (/qs3/pubsys2/xml/en/procedures/89/89-003-999.html)
Procedure Title	Rocker Levers - Overview
Update Reason	
Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-004-999 (/qs3/pubsys2/xml/en/procedures/89/89-004-999.html)
Procedure Title	Cam Followers/Tappets - Overview
Update Reason	
Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-005-999 (/qs3/pubsys2/xml/en/procedures/89/89-005-999.html)
Procedure Title	Fuel System - Overview
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-006-999 (/qs3/pubsys2/xml/en/procedures/89/89-006-999.html)
Procedure Title	Injectors and Fuel Lines - Overview
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-007-999 (/qs3/pubsys2/xml/en/procedures/89/89-007-999.html)
Procedure Title	Lubricating Oil System - Overview
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-008-999 (/qs3/pubsys2/xml/en/procedures/89/89-008-999.html)
Procedure Title	Cooling System - Overview
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-010-999 (/qs3/pubsys2/xml/en/procedures/89/89-010-999.html)
Procedure Title	Air Intake System - Overview
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-011-999 (/qs3/pubsys2/xml/en/procedures/89/89-011-999.html)
Procedure Title	Exhaust System - Overview
Update Reason	

Date	08-Jul-2003
Section	Section F - Familiarization
Procedure Number	89-016-999 (/qs3/pubsys2/xml/en/procedures/89/89-016-999.html)
Procedure Title	Mounting Adaptations - Overview
Update Reason	
Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t033-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t033-tr.html)
Procedure Title	Engine Acceleration or Response Poor
Update Reason	
Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t041-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t041-tr.html)
Procedure Title	Engine Decelerates Slowly
Update Reason	
Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t043-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t043-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (Exhaust Smoke)
Update Reason	
Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t044-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t044-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	

Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t057-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t057-tr.html)
Procedure Title	Engine Power Output Low
Update Reason	

Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t061-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t061-tr.html)
Procedure Title	Engine Runs Rough at Idle
Update Reason	

Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t062-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t062-tr.html)
Procedure Title	Engine Runs Rough or Misfires
Update Reason	

Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t066-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t066-tr.html)
Procedure Title	Engine Speed Surges at Low or High Idle
Update Reason	

Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t080-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t080-tr.html)
Procedure Title	Engine Will Not Reach Rated Speed (RPM)
Update Reason	

Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t087-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	
Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t116-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t116-tr.html)
Procedure Title	Smoke, Black - Excessive
Update Reason	
Date	16-Jul-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t118-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t118-tr.html)
Procedure Title	Smoke, White - Excessive
Update Reason	
Date	28-Jul-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-018 (/qs3/pubsys2/xml/en/procedures/89/89-001-018.html)
Procedure Title	Crankshaft Gear, Front (Crankshaft Installed)
Update Reason	
Date	28-Jul-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-031 (/qs3/pubsys2/xml/en/procedures/89/89-001-031.html)
Procedure Title	Gear Cover, Front
Update Reason	

Date	28-Jul-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-052 (/qs3/pubsys2/xml/en/procedures/89/89-001-052.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	
Date	28-Jul-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-054 (/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	
Date	28-Jul-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-055 (/qs3/pubsys2/xml/en/procedures/89/89-001-055.html)
Procedure Title	Gear Train Backlash, Front
Update Reason	
Date	28-Jul-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-089 (/qs3/pubsys2/xml/en/procedures/89/89-001-089.html)
Procedure Title	Block Stiffener Plate
Update Reason	
Date	28-Jul-2003
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-049 (/qs3/pubsys2/xml/en/procedures/89/89-006-049.html)
Procedure Title	Fuel Control Valve Body Assembly
Update Reason	

Date	28-Jul-2003
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	89-007-028 (/qs3/pubsys2/xml/en/procedures/89/89-007-028.html)
Procedure Title	Lubricating Oil Pressure Gauge
Update Reason	
Date	28-Jul-2003
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-022-001_10 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_10.html)
Procedure Title	Service Tools
Update Reason	
Date	28-Jul-2003
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-033 (/qs3/pubsys2/xml/en/procedures/89/89-010-033.html)
Procedure Title	Turbocharger
Update Reason	
Date	28-Jul-2003
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-019 (/qs3/pubsys2/xml/en/procedures/89/89-013-019.html)
Procedure Title	Starter Solenoid
Update Reason	
Date	28-Jul-2003
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-016-002 (/qs3/pubsys2/xml/en/procedures/89/89-016-002.html)
Procedure Title	Engine Support Bracket, Front
Update Reason	

Date	04-Aug-2003
Section	Section F - Familiarization
Procedure Number	89-009-999 (/qs3/pubsys2/xml/en/procedures/89/89-009-999.html)
Procedure Title	Drive Untis - Overview
Update Reason	
Date	04-Aug-2003
Section	Section F - Familiarization
Procedure Number	89-013-999 (/qs3/pubsys2/xml/en/procedures/89/89-013-999.html)
Procedure Title	Electrical Equipment - Overview
Update Reason	
Date	04-Aug-2003
Section	Section F - Familiarization
Procedure Number	89-014-999 (/qs3/pubsys2/xml/en/procedures/89/89-014-999.html)
Procedure Title	Engine Testing - Overview
Update Reason	
Date	04-Aug-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-022-001_01 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_01.html)
Procedure Title	Service Tools
Update Reason	
Date	04-Aug-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-001 (/qs3/pubsys2/xml/en/procedures/89/89-001-001.html)
Procedure Title	Alternator Drive Seal
Update Reason	

Date	04-Aug-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-008 (/qs3/pubsys2/xml/en/procedures/89/89-001-008.html)
Procedure Title	Camshaft
Update Reason	
Date	04-Aug-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-012 (/qs3/pubsys2/xml/en/procedures/89/89-001-012.html)
Procedure Title	Camshaft Gear (Camshaft Installed)
Update Reason	
Date	04-Aug-2003
Section	Section 17 - Miscellaneous - Group 17
Procedure Number	89-017-007 (/qs3/pubsys2/xml/en/procedures/89/89-017-007.html)
Procedure Title	Pipe Plug
Update Reason	
Date	04-Aug-2003
Section	Section 17 - Miscellaneous - Group 17
Procedure Number	89-017-011 (/qs3/pubsys2/xml/en/procedures/89/89-017-011.html)
Procedure Title	Straight Thread Plug
Update Reason	
Date	26-Aug-2003
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	99-013-009 (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
Procedure Title	Battery Cables and Connections
Update Reason	

Date	04-Sep-2003
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-017 (/qs3/pubsys2/xml/en/procedures/89/89-013-017.html)
Procedure Title	Starter Magnetic Switch
Update Reason	
Date	04-Sep-2003
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-018 (/qs3/pubsys2/xml/en/procedures/89/89-013-018.html)
Procedure Title	Starter Switch
Update Reason	
Date	15-Sep-2003
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	89-022-001_03 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_03.html)
Procedure Title	Service Tools
Update Reason	
Date	15-Sep-2003
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-022-001_06 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_06.html)
Procedure Title	Service Tools
Update Reason	
Date	15-Sep-2003
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-003 (/qs3/pubsys2/xml/en/procedures/89/89-006-003.html)
Procedure Title	Air in Fuel
Update Reason	

Date	15-Sep-2003
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	89-007-007 (/qs3/pubsys2/xml/en/procedures/89/89-007-007.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	
Date	23-Sep-2003
Section	front
Procedure Number	4021375-titlepage (/qs3/pubsys2/xml/en/manual/4021375/4021375-titlepage.html)
Procedure Title	titlepage
Update Reason	
Date	23-Sep-2003
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-022-001_02 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_02.html)
Procedure Title	Service Tools
Update Reason	
Date	23-Sep-2003
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-012 (/qs3/pubsys2/xml/en/procedures/89/89-006-012.html)
Procedure Title	Fuel Drain Line Restriction
Update Reason	
Date	23-Sep-2003
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-015 (/qs3/pubsys2/xml/en/procedures/89/89-006-015.html)
Procedure Title	Fuel Filter (Spin-On Type)
Update Reason	

Date	13-Nov-2003
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-026 (/qs3/pubsys2/xml/en/procedures/89/89-006-026.html)
Procedure Title	Injector
Update Reason	
Date	19-Dec-2003
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-014 (/qs3/pubsys2/xml/en/procedures/89/89-001-014.html)
Procedure Title	Connecting Rod
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	

Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	
Date	29-Mar-2004
Section	Section 14 - Engine Testing - Group 14
Procedure Number	00-014-012 (/qs3/pubsys2/xml/en/procedures/00/00-014-012.html)
Procedure Title	Engine Compression
Update Reason	
Date	29-Sep-2004
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	
Date	25-Oct-2004
Section	back
Procedure Number	4021375-backcover (/qs3/pubsys2/xml/en/manual/4021375/4021375-backcover.html)
Procedure Title	Backcover
Update Reason	

Date	22-May-2006
Section	front
Procedure Number	tr-frwd (/qs3/pubsys2/xml/en/manual/preface/tr-frwd.html)
Procedure Title	foreword
Update Reason	
Date	26-Jun-2006
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-014-005 (/qs3/pubsys2/xml/en/procedures/89/89-014-005.html)
Procedure Title	Engine Testing (Engine Dynamometer)
Update Reason	
Date	26-Jun-2006
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-014-006 (/qs3/pubsys2/xml/en/procedures/89/89-014-006.html)
Procedure Title	Engine Run-in (Engine Dynamometer)
Update Reason	
Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	21-Jul-2006
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-022-001_14 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_14.html)
Procedure Title	Service Tools
Update Reason	

Date	16-May-2007
Section	Section V - Specifications
Procedure Number	89-018-016 (/qs3/pubsys2/xml/en/procedures/89/89-018-016.html)
Procedure Title	Fuel System
Update Reason	
Date	16-May-2007
Section	Section V - Specifications
Procedure Number	89-018-017 (/qs3/pubsys2/xml/en/procedures/89/89-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	
Date	16-May-2007
Section	Section V - Specifications
Procedure Number	89-018-018 (/qs3/pubsys2/xml/en/procedures/89/89-018-018.html)
Procedure Title	Cooling System
Update Reason	
Date	16-May-2007
Section	Section V - Specifications
Procedure Number	89-018-019 (/qs3/pubsys2/xml/en/procedures/89/89-018-019.html)
Procedure Title	Air Intake System
Update Reason	
Date	16-May-2007
Section	Section V - Specifications
Procedure Number	89-018-020 (/qs3/pubsys2/xml/en/procedures/89/89-018-020.html)
Procedure Title	Exhaust System
Update Reason	

Date	16-May-2007
Section	Section V - Specifications
Procedure Number	89-018-021 (/qs3/pubsys2/xml/en/procedures/89/89-018-021.html)
Procedure Title	Electrical System
Update Reason	
Date	28-Jun-2007
Section	Section 8 - Cooling System - Group 08
Procedure Number	89-008-018-tr (/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	
Date	23-Jul-2007
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t021-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t021-tr.html)
Procedure Title	Coolant Loss - Internal
Update Reason	
Date	23-Jul-2007
Section	Section TS - Troubleshooting Symptoms
Procedure Number	89-t00-004 (/qs3/pubsys2/xml/en/procedures/89/89-t00-004.html)
Procedure Title	Troubleshooting Overview
Update Reason	
Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	

Date	03-Aug-2007
Section	Section 8 - Cooling System - Group 08
Procedure Number	89-022-001_08-tr (/qs3/pubsys2/xml/en/procedures/89/89-022-001_08-tr.html)
Procedure Title	Service Tools
Update Reason	
Date	05-Dec-2007
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t027-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t027-tr.html)
Procedure Title	Crankcase Gases (Blowby) Excessive
Update Reason	
Date	06-Feb-2008
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-016-006 (/qs3/pubsys2/xml/en/procedures/89/89-016-006.html)
Procedure Title	Flywheel Housing
Update Reason	
Date	24-Sep-2008
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	89-003-006 (/qs3/pubsys2/xml/en/procedures/89/89-003-006.html)
Procedure Title	Overhead Set (OBC)
Update Reason	
Date	24-Sep-2008
Section	Section L - Service Literature
Procedure Number	89-205-001-tr (/qs3/pubsys2/xml/en/procedures/89/89-205-001-tr.html)
Procedure Title	Additional Service Literature
Update Reason	

Date	24-Sep-2008
Section	Section V - Specifications
Procedure Number	89-018-015 (/qs3/pubsys2/xml/en/procedures/89/89-018-015.html)
Procedure Title	General Engine
Update Reason	
Date	13-Nov-2008
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	00-016-010 (/qs3/pubsys2/xml/en/procedures/00/00-016-010.html)
Procedure Title	Engine Mounts
Update Reason	
Date	20-Jan-2009
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-002-004 (/qs3/pubsys2/xml/en/procedures/89/89-002-004.html)
Procedure Title	Cylinder Head
Update Reason	
Date	26-Mar-2009
Section	Section TT - Troubleshooting Symptoms
Procedure Number	105-t02-1005a (/qs3/pubsys2/xml/en/procedures/105/105-t02-1005a.html)
Procedure Title	Vibration Excessive
Update Reason	
Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	

Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	27-Jul-2009
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	89-012-014 (/qs3/pubsys2/xml/en/procedures/89/89-012-014.html)
Procedure Title	Air Compressor
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	
Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	

Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	25-Feb-2010
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t102-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	24-Mar-2010
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-016-005 (/qs3/pubsys2/xml/en/procedures/89/89-016-005.html)
Procedure Title	Flywheel
Update Reason	
Date	25-May-2010
Section	Section i - Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	

Date	04-OCT-2010
Section	Section 01 - Cylinder Block
Procedure Number	89-001-054 (/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Added General Information Step
Date	23-Sep-2003
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-002-004-tr (/qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Updated technical content of Vacuum Test section.
Date	2-Nov-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-026 (/qs3/pubsys2/xml/en/procedures/89/89-001-026.html)
Procedure Title	Cylinder Block
Update Reason	Updated procedure to include main bearing bore reuse guidelines.
Date	30-Nov-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-007 (/qs3/pubsys2/xml/en/procedures/89/89-001-007.html)
Procedure Title	Bearings, Thrust
Update Reason	Changed technical information in table of Inspect for Reuse step.

Date	10-Dec-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-006 (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)
Procedure Title	Bearings, Main
Update Reason	Added General Information to procedure, updated graphic, and technical information in table in the Install step.

Date	21-Dec-2010
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-016-tr (/qs3/pubsys2/xml/en/procedures/89/89-001-016-tr.html)
Procedure Title	Crankshaft
Update Reason	Removed Maintenance Check step from procedure and updated technical information for measuring the outside diameter.

Date	10-Jan-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-008 (/qs3/pubsys2/xml/en/procedures/89/89-001-008.html)
Procedure Title	Camshaft
Update Reason	Removed statement about injector timing in the Preparatory and Finish sections.

Date	10-Jan-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-010 (/qs3/pubsys2/xml/en/procedures/89/89-001-010.html)
Procedure Title	Camshaft Bushings
Update Reason	Removed statement about injector timing in the Preparatory and Finish sections.

Date	10-Jan-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-012 (/qs3/pubsys2/xml/en/procedures/89/89-001-012.html)
Procedure Title	Camshaft Gear (Camshaft Installed)
Update Reason	Removed statement about injector timing in the Preparatory and Finish sections.

Date	10-Jan-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-013 (/qs3/pubsys2/xml/en/procedures/89/89-001-013.html)
Procedure Title	Camshaft Gear (Camshaft Removed)
Update Reason	Removed statement about injector timing in the Preparatory and Finish sections.

Date	10-Jan-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-056 (/qs3/pubsys2/xml/en/procedures/89/89-001-056.html)
Procedure Title	Camshaft Thrust Bearing
Update Reason	Removed statement about injector timing in the Preparatory and Finish sections.

Date	03-May-2011
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-016-006 (/qs3/pubsys2/xml/en/procedures/89/89-016-006.html)
Procedure Title	Flywheel Housing
Update Reason	Added in Install step a paragraph to install rear engine support brackets, washers, and bolts and torque 360 N•m [265 ft-lb].

Date	03-May-2011
Section	Section V - Maintenance Specifications
Procedure Number	89-018-017 (/qs3/pubsys2/xml/en/procedures/89/89-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Added Oil Sump Capacity table.
Date	07-Jun-2011
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	Removed reference to oversized cylinder liners as these are not available on the QSK23 engine.
Date	23-Nov-2011
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-023 (/qs3/pubsys2/xml/en/procedures/89/89-010-023.html)
Procedure Title	Air Intake Manifold
Update Reason	Added steps for Remove and Install air crossover connection.
Date	23-Nov-2011
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-072-tr (/qs3/pubsys2/xml/en/procedures/89/89-010-072-tr.html)
Procedure Title	Air Intake Manifold
Update Reason	

Date	03-Feb-2012
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-072-tr (/qs3/pubsys2/xml/en/procedures/89/89-010-072-tr.html)
Procedure Title	Air Intake Manifold
Update Reason	Updated the procedure with current informtion and added it to some manual outlines.
Date	06-Feb-2012
Section	Section 8 - Cooling System - Group 08
Procedure Number	89-008-062-tr (/qs3/pubsys2/xml/en/procedures/89/89-008-062-tr.html)
Procedure Title	Water Pump
Update Reason	Updated procedure by adding Assemble and Disassemble sections.
Date	21-Mar-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-024 (/qs3/pubsys2/xml/en/procedures/89/89-001-024.html)
Procedure Title	Cylinder Block
Update Reason	Updated seal removal/installation tool, Part Number 5298540, and instructions.
Date	21-Mar-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-024 (/qs3/pubsys2/xml/en/procedures/89/89-001-024.html)
Procedure Title	Cylinder Block
Update Reason	Added seal removal/installation tool, Part Number 5298540.

Date	02-Apr-2012
Section	Section 9 - Drive Units - Group 09
Procedure Number	89-009-009 (/qs3/pubsys2/xml/en/procedures/89/89-009-009.html)
Procedure Title	Alternator Drive Gear and Shaft
Update Reason	Added assemble and disassemble sections to the procedure.
Date	29-May-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-006 (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)
Procedure Title	Bearings, Main
Update Reason	Added content for oversize bearing specifications.
Date	29-May-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-054 (/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Added content for oversize bearing specifications.
Date	11-Jun-2012
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021375-t021-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t021-tr.html)
Procedure Title	Coolant Loss - Internal
Update Reason	Add Step, Cause: Cylinder block is cracked or porous, Correction: Perform the pressure test procedure (008-018) with tracer dye in the coolant. Inspect the cylinder block. Refer to Procedure 001-026 in Section 1.

Date	25-Jun-2012
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-028-tr (/qs3/pubsys2/xml/en/procedures/89/89-001-028-tr.html)
Procedure Title	Cylinder Liner
Update Reason	Updated the Remove step to add instruction that liner seals must be discarded if removed from the liner.
Date	20-Aug-2012
Section	Section TS - Troubleshooting System
Procedure Number	4021375-t102-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Changed first step to include inspection of OEM components for external oil leaks.
Date	05-Jun-2013
Section	Section 8 - Cooling System - Group 08
Procedure Number	89-022-001_08-tr (/qs3/pubsys2/xml/en/procedures/89/89-022-001_08-tr.html)
Procedure Title	Srvce Tools
Update Reason	Added water pump replacer kit.
Date	26-Aug-2013
Section	Section 14 - Engine Testing - Group 14
Procedure Number	00-014-012 (/qs3/pubsys2/xml/en/procedures/00/00-014-012.html)
Procedure Title	Engine Compression
Update Reason	Updated Preparatory Step and Finishing Step procedure links.

Date	10-Sep-2013
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-001 (/qs3/pubsys2/xml/en/procedures/89/89-013-001.html)
Procedure Title	Alternator
Update Reason	Removed the belt tension specifications due to multiple specification. locations.

Date	10-Sep-2013
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-005 (/qs3/pubsys2/xml/en/procedures/89/89-013-005.html)
Procedure Title	Drive Belt, Alternator
Update Reason	Updated the proper belt tension specification for the different belt types.

Date	05-Nov-2013
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	102-016-008 (/qs3/pubsys2/xml/en/procedures/102/102-016-008.html)
Procedure Title	Flywheel Ring Gear
Update Reason	Added procedure to manual.

Date	14-Jan-2014
Section	Section TS - Troubleshooting System
Procedure Number	4021375-t102-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t102-tr.html)
Procedure Title	Lubricating Oil Consumption Excessive
Update Reason	Added step three to TS tree. Verifying original equipment manufacturer (OEM) lubricating oil system components.

Date	29-Apr-2013
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-002-004-tr (/qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Made reference to injector protrusion and the proper sealing ring are made.

Date	29-Apr-2014
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-026-tr (/qs3/pubsys2/xml/en/procedures/89/89-006-026-tr.html)
Procedure Title	Injector
Update Reason	Revision of Inspect for Reuse section, and install section with requirements to replace the injector sealing ring (mechanically actuated injector only), if removed.

Date	29-Apr-2013
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-002-022 (/qs3/pubsys2/xml/en/procedures/89/89-002-022.html)
Procedure Title	Injector Protrusion
Update Reason	Added procedure to outline.

Date	09- May-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-036 (/qs3/pubsys2/xml/en/procedures/89/89-001-036.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	A stepblock and graphic is added illustrating the three capscrews, the shaft, the camshaft idler gear assembly, and the rear thrust bearing.

Date	09-May-2014
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-005 (/qs3/pubsys2/xml/en/procedures/89/89-013-005.html)
Procedure Title	Drive Belt, Alternator
Update Reason	Added 'belt deflection method' for a cogged belt, when setting tension in the Adjust step.
Date	05-Jun-2014
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-002-004-tr (/qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Corrected valve guide height specification which was determined to be 54.0 +/-0.25 mm based on the part drawing information and verified with dimensions of the tool used for valve guide insertion.
Date	28-Jul-2014
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-022-001_13 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_13.html)
Procedure Title	Service Tools
Update Reason	Added Tension Gauge, Part Number 5299312.
Date	08-Oct-2014
Section	Section TS - Troubleshooting Symptoms - Group t00
Procedure Number	4021375-t107-tr (/qs3/pubsys2/xml/en/manual/4021375/4021375-t107-tr.html)
Procedure Title	Lubricating Oil Temperature Above Specification
Update Reason	Updated TS tree with correct wording for field service on lubricating oil above or below specification.

Date	20-Oct-2014
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	89-007-017 (/qs3/pubsys2/xml/en/procedures/89/89-007-017.html)
Procedure Title	Lubricating Oil Filter Head (Remote-Mounted)
Update Reason	Added to manual.
Date	11-Nov-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-006 (/qs3/pubsys2/xml/en/procedures/89/89-001-006.html)
Procedure Title	Bearings, Main
Update Reason	Updated the Main Bearing bearing thickness measurements.
Date	11-Nov-2014
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-054 (/qs3/pubsys2/xml/en/procedures/89/89-001-054.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Updated the Connecting Rod bearing thickness measurements.
Date	25-Nov-2014
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	89-012-003 (/qs3/pubsys2/xml/en/procedures/89/89-012-003.html)
Procedure Title	Air Compressor Carbon Buildup
Update Reason	Added to manual.

Date	25-Nov-2014
Section	Section 12 - Compressed Air System - Group 12
Procedure Number	89-012-013 (/qs3/pubsys2/xml/en/procedures/89/89-012-013.html)
Procedure Title	Air Compressor Unloader and Valve Assembly
Update Reason	Added to manual.
Date	25-Nov-2014
Section	Section 5 - Fuel System - Group 05
Procedure Number	89-005-016 (/qs3/pubsys2/xml/en/procedures/89/89-005-016.html)
Procedure Title	Fuel Pump
Update Reason	Corrected pressure values for the fuel pump.
Date	Not Available
Section	front
Procedure Number	3666013-toc (/qs3/pubsys2/xml/en/manual/3666013/3666013-toc.html)
Procedure Title	Table of Contents
Update Reason	
Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-004 (/qs3/pubsys2/xml/en/procedures/99/99-204-004.html)
Procedure Title	Symbols
Update Reason	
Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-005 (/qs3/pubsys2/xml/en/procedures/99/99-204-005.html)
Procedure Title	Illustrations
Update Reason	

Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-103-999 (/qs3/pubsys2/xml/en/procedures/99/99-103-999.html)
Procedure Title	Daily Maintenance Procedures
Update Reason	
Date	14-Mar-2002
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-006-043 (/qs3/pubsys2/xml/en/procedures/99/99-006-043.html)
Procedure Title	Fuel-Water Separator
Update Reason	
Date	14-Mar-2002
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedures
Update Reason	
Date	14-Mar-2002
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	99-010-058 (/qs3/pubsys2/xml/en/procedures/99/99-010-058.html)
Procedure Title	Air Intake Piping
Update Reason	

Date	14-Mar-2002
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	99-012-021 (/qs3/pubsys2/xml/en/procedures/99/99-012-021.html)
Procedure Title	Air Tanks and Reservoirs
Update Reason	
Date	14-Mar-2002
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedures - Overview
Update Reason	
Date	14-Mar-2002
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	99-003-018 (/qs3/pubsys2/xml/en/procedures/99/99-003-018.html)
Procedure Title	Crankcase Breather
Update Reason	
Date	14-Mar-2002
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedures - Overview
Update Reason	

Date	14-Mar-2002
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	99-999-999 (/qs3/pubsys2/xml/en/procedures/99/99-999-999.html)
Procedure Title	Maintenance Procedure
Update Reason	
Date	14-Mar-2002
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	99-012-015 (/qs3/pubsys2/xml/en/procedures/99/99-012-015.html)
Procedure Title	Air Compressor Discharger Lines
Update Reason	
Date	03-Apr-2002
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-002 (/qs3/pubsys2/xml/en/procedures/99/99-t00-002.html)
Procedure Title	Troubleshooting Symptom Charts
Update Reason	
Date	08-Oct-2002
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	99-008-040 (/qs3/pubsys2/xml/en/procedures/99/99-008-040.html)
Procedure Title	Fan, Cooling
Update Reason	

Date	31-Oct-2002
Section	Section 1 - Operating Instructions
Procedure Number	00-101-014 (/qs3/pubsys2/xml/en/procedures/00/00-101-014.html)
Procedure Title	Normal Starting Procedure
Update Reason	
Date	13-Feb-2003
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-010-015 (/qs3/pubsys2/xml/en/procedures/99/99-010-015.html)
Procedure Title	Air Cleaner Precleaner
Update Reason	
Date	21-Apr-2003
Section	Section D - System Diagrams
Procedure Number	00-200-999 (/qs3/pubsys2/xml/en/procedures/00/00-200-999.html)
Procedure Title	System Diagrams - Overview
Update Reason	
Date	26-Aug-2003
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	99-013-007 (/qs3/pubsys2/xml/en/procedures/99/99-013-007.html)
Procedure Title	Batteries
Update Reason	

Date	26-Aug-2003
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	99-013-009 (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
Procedure Title	Battery Cables and Connections
Update Reason	
Date	29-Jan-2004
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-008-067 (/qs3/pubsys2/xml/en/procedures/99/99-008-067.html)
Procedure Title	Seawater Strainer
Update Reason	
Date	02-Mar-2004
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	102-010-061 (/qs3/pubsys2/xml/en/procedures/102/102-010-061.html)
Procedure Title	Air Cleaner Element, Single Heavy-Duty Dry Type
Update Reason	
Date	02-Mar-2004
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	102-010-062 (/qs3/pubsys2/xml/en/procedures/102/102-010-062.html)
Procedure Title	Air Cleaner Element, Dual Heavy Duty Dry Type
Update Reason	

Date	02-Mar-2004
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	102-010-063 (/qs3/pubsys2/xml/en/procedures/102/102-010-063.html)
Procedure Title	Air Cleaner Element, Cartridge Type
Update Reason	
Date	02-Mar-2004
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	102-012-022 (/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
Procedure Title	Air Starting Motor
Update Reason	
Date	02-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	102-018-004 (/qs3/pubsys2/xml/en/procedures/102/102-018-004.html)
Procedure Title	Coolant Recommendations and Specifications
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	

Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	
Date	03-Mar-2004
Section	Section V - Maintenance Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	
Date	06-Oct-2004
Section	Section E - Engine Identification
Procedure Number	18-100-001 (/qs3/pubsys2/xml/en/procedures/18/18-100-001.html)
Procedure Title	Engine Identification
Update Reason	
Date	19-Oct-2004
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	18-003-001 (/qs3/pubsys2/xml/en/procedures/18/18-003-001.html)
Procedure Title	Crankcase Breather, External
Update Reason	

Date	28-Sep-2005
Section	Section SS - Seasonal Storage (Marine Only)
Procedure Number	99-000-014 (/qs3/pubsys2/xml/en/procedures/99/99-000-014.html)
Procedure Title	Seasonal Storage
Update Reason	
Date	15-Feb-2006
Section	front
Procedure Number	refno (/qs3/pubsys2/xml/en/procedures//refno.html)
Procedure Title	Important Reference Numbers
Update Reason	
Date	20-Apr-2006
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	00-008-059 (/qs3/pubsys2/xml/en/procedures/00/00-008-059.html)
Procedure Title	Zinc Anode
Update Reason	
Date	04-May-2006
Section	Section i - Introduction
Procedure Number	99-204-001 (/qs3/pubsys2/xml/en/procedures/99/99-204-001.html)
Procedure Title	To the Owner and Operator
Update Reason	
Date	22-May-2006
Section	front
Procedure Number	om-frwd (/qs3/pubsys2/xml/en/manual/preface/om-frwd.html)
Procedure Title	Foreword
Update Reason	

Date	31-May-2006
Section	Section S - Service Assistance
Procedure Number	99-202-005 (/qs3/pubsys2/xml/en/procedures/99/99-202-005.html)
Procedure Title	Division and Regional Offices
Update Reason	
Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t024-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t024-om.html)
Procedure Title	Coolant Temperature is Below Normal
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t026-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t026-om.html)
Procedure Title	Coolant Temperature Above Normal
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t026-155-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t026-155-om.html)
Procedure Title	Coolant Temperature Above Normal (Marine)
Update Reason	

Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t043-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t043-om.html)
Procedure Title	Engine Hard to Start or Will Not Start (Exhaust Smoke)
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t044-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t044-om.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t061-011-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t061-011-om.html)
Procedure Title	Engine Runs Rough at Idle, Warm Engine
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t072-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t072-om.html)
Procedure Title	Engine Starts But Will Not Keep Running
Update Reason	

Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t077-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t077-om.html)
Procedure Title	Engine Will Not Crank or Cranks Slowly (Air Starter)
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t078-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t078-om.html)
Procedure Title	Engine Will Not Crank
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t080-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t080-om.html)
Procedure Title	Engine Will Not Reach Rated Speed (RPM)
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t081-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t081-om.html)
Procedure Title	Engine Will Not Shut Off
Update Reason	
Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t081-006-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t081-006-om.html)
Procedure Title	Exhaust Smoke Excessive Under Load
Update Reason	

Date	14-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t105-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t105-om.html)
Procedure Title	Lubricating Oil Pressure Low
Update Reason	
Date	14-Jul-2006
Section	back
Procedure Number	3666013-backcover (/qs3/pubsys2/xml/en/manual/3666013/3666013-backcover.html)
Procedure Title	Backcover
Update Reason	
Date	19-Jul-2006
Section	Section TS - Troubleshooting Symptoms
Procedure Number	3666013-t057-om (/qs3/pubsys2/xml/en/manual/3666013/3666013-t057-om.html)
Procedure Title	Engine Power Output Low
Update Reason	
Date	24-Jul-2006
Section	Section S - Service Assistance
Procedure Number	99-202-008 (/qs3/pubsys2/xml/en/procedures/99/99-202-008.html)
Procedure Title	Distributors - International
Update Reason	
Date	28-Jul-2006
Section	Section 1 - Operating Instructions
Procedure Number	18-101-019 (/qs3/pubsys2/xml/en/procedures/18/18-101-019.html)
Procedure Title	Step Timing Control (STC)
Update Reason	

Date	28-Jul-2006
Section	Section 2 - Maintenance Guidelines
Procedure Number	18-102-004 (/qs3/pubsys2/xml/en/procedures/18/18-102-004.html)
Procedure Title	Tool Requirements
Update Reason	

Date	28-Jul-2006
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	18-103-001 (/qs3/pubsys2/xml/en/procedures/18/18-103-001.html)
Procedure Title	Engine Operation Report
Update Reason	

Date	28-Jul-2006
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	18-103-004 (/qs3/pubsys2/xml/en/procedures/18/18-103-004.html)
Procedure Title	Unusual Engine Noise
Update Reason	

Date	28-Jul-2006
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	18-010-086 (/qs3/pubsys2/xml/en/procedures/18/18-010-086.html)
Procedure Title	Air Cleaner Restriction Indicator
Update Reason	

Date	28-Jul-2006
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	18-010-014 (/qs3/pubsys2/xml/en/procedures/18/18-010-014.html)
Procedure Title	Air Cleaner Element
Update Reason	
Date	28-Jul-2006
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	18-006-015-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-006-015-om-ind.html)
Procedure Title	Fuel Filter (Spin-On)
Update Reason	
Date	28-Jul-2006
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	18-007-002 (/qs3/pubsys2/xml/en/procedures/18/18-007-002.html)
Procedure Title	Lubricating Oil and Filters
Update Reason	
Date	28-Jul-2006
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	18-007-013-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-007-013-om-ind.html)
Procedure Title	Lubricating Oil Filter (Spin-On)
Update Reason	

Date	28-Jul-2006
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	18-008-006-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-006-om-ind.html)
Procedure Title	Coolant Filter
Update Reason	
Date	28-Jul-2006
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	18-012-001 (/qs3/pubsys2/xml/en/procedures/18/18-012-001.html)
Procedure Title	Air Compressor Air Cleaner Element
Update Reason	
Date	28-Jul-2006
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	18-008-002-om-rep (/qs3/pubsys2/xml/en/procedures/18/18-008-002-om-rep.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	
Date	28-Jul-2006
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	18-003-006-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-003-006-om-ind.html)
Procedure Title	Overhead Set (OBC)
Update Reason	

Date	28-Jul-2006
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	18-003-007-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-003-007-om-ind.html)
Procedure Title	Overhead Set (Dial Indicator)
Update Reason	
Date	28-Jul-2006
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	18-008-045-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-045-om-ind.html)
Procedure Title	Radiator Hoses
Update Reason	
Date	28-Jul-2006
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	18-008-029-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-029-om-ind.html)
Procedure Title	Fan Drive Idler Arm Assembly
Update Reason	
Date	28-Jul-2006
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	18-016-002-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-016-002-om-ind.html)
Procedure Title	Engine Support Bracket, Front
Update Reason	

Date	28-Jul-2006
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-008-002-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-002-om-ind.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-005-016-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-005-016-om-ind.html)
Procedure Title	Fuel Pump
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-006-026-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-006-026-om-ind.html)
Procedure Title	Injectors
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-008-018-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-018-om-ind.html)
Procedure Title	Cooling System
Update Reason	

Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-008-036-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-036-om-ind.html)
Procedure Title	Fan Hub, Belt Driven
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-008-037-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-037-om-ind.html)
Procedure Title	Fan Hub, Gear Driven
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-008-030-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-030-om-ind.html)
Procedure Title	Fan Drive Idler Pulley Assembly
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-008-062-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-062-om-ind.html)
Procedure Title	Water Pump
Update Reason	

Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-010-033-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-010-033-om-ind.html)
Procedure Title	Turbocharger
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-012-014-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-012-014-om-ind.html)
Procedure Title	Air Compressor
Update Reason	
Date	31-Jul-2006
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-012-013 (/qs3/pubsys2/xml/en/procedures/18/18-012-013.html)
Procedure Title	Air Compressor Unloader and Valve Assembly
Update Reason	
Date	31-Jul-2006
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-013-005-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-013-005-om-ind.html)
Procedure Title	Drive Belt, Alternator
Update Reason	

Date	31-Jul-2006
Section	Section L - Service Literature
Procedure Number	18-205-001-om (/qs3/pubsys2/xml/en/procedures/18/18-205-001-om.html)
Procedure Title	Additional Service Literature
Update Reason	
Date	31-Jul-2006
Section	Section V - Maintenance Specifications
Procedure Number	18-018-024 (/qs3/pubsys2/xml/en/procedures/18/18-018-024.html)
Procedure Title	Cummins/Fleetguard® Filter Specifications
Update Reason	
Date	31-Jul-2006
Section	Section V - Maintenance Specifications
Procedure Number	18-018-006 (/qs3/pubsys2/xml/en/procedures/18/18-018-006.html)
Procedure Title	Engine Component Torque Values
Update Reason	
Date	17-Aug-2006
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-003-011-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-003-011-om-ind.html)
Procedure Title	Rocker Lever Cover
Update Reason	
Date	03-Oct-2006
Section	Section E - Engine Identification
Procedure Number	18-100-002 (/qs3/pubsys2/xml/en/procedures/18/18-100-002.html)
Procedure Title	Engine Diagrams
Update Reason	

Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	18-200-001 (/qs3/pubsys2/xml/en/procedures/18/18-200-001.html)
Procedure Title	Flow Diagram, Fuel System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	18-200-002 (/qs3/pubsys2/xml/en/procedures/18/18-200-002.html)
Procedure Title	Flow Diagram, Lubricating Oil System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	18-200-003 (/qs3/pubsys2/xml/en/procedures/18/18-200-003.html)
Procedure Title	Flow Diagram, Cooling System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	18-200-004 (/qs3/pubsys2/xml/en/procedures/18/18-200-004.html)
Procedure Title	Flow Diagram, Air Intake System
Update Reason	
Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	18-200-005 (/qs3/pubsys2/xml/en/procedures/18/18-200-005.html)
Procedure Title	Flow Diagram, Exhaust System
Update Reason	

Date	03-Oct-2006
Section	Section D - System Diagrams
Procedure Number	18-200-006 (/qs3/pubsys2/xml/en/procedures/18/18-200-006.html)
Procedure Title	Flow Diagram, Compressed Air System
Update Reason	
Date	10-Oct-2006
Section	front
Procedure Number	3666013-titlepage (/qs3/pubsys2/xml/en/manual/3666013/3666013-titlepage.html)
Procedure Title	Titlepage
Update Reason	
Date	10-Oct-2006
Section	Section 2 - Maintenance Guidelines
Procedure Number	18-102-002-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-102-002-om-ind.html)
Procedure Title	Maintenance Schedule
Update Reason	
Date	31-Jan-2007
Section	Section V - Maintenance Specifications
Procedure Number	102-018-002 (/qs3/pubsys2/xml/en/procedures/102/102-018-002.html)
Procedure Title	Fuel Recommendations and Specifications
Update Reason	
Date	23-Apr-2007
Section	Section W - Warranty
Procedure Number	3381322 (/qs3/pubsys2/xml/en/warranty/3381322.html)
Procedure Title	All Engines International Industrial (Off-Highway)
Update Reason	

Date	08-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	102-018-003 (/qs3/pubsys2/xml/en/procedures/102/102-018-003.html)
Procedure Title	Lubricating Oil Recommendations and Specifications
Update Reason	
Date	16-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-015 (/qs3/pubsys2/xml/en/procedures/18/18-018-015.html)
Procedure Title	General Specifications
Update Reason	
Date	16-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-018 (/qs3/pubsys2/xml/en/procedures/18/18-018-018.html)
Procedure Title	Cooling System
Update Reason	
Date	16-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-019 (/qs3/pubsys2/xml/en/procedures/18/18-018-019.html)
Procedure Title	Air Intake System
Update Reason	
Date	16-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-020 (/qs3/pubsys2/xml/en/procedures/18/18-018-020.html)
Procedure Title	Exhaust System
Update Reason	

Date	16-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-021 (/qs3/pubsys2/xml/en/procedures/18/18-018-021.html)
Procedure Title	Electrical System
Update Reason	
Date	16-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-022 (/qs3/pubsys2/xml/en/procedures/18/18-018-022.html)
Procedure Title	Compressed Air System
Update Reason	
Date	24-May-2007
Section	Section V - Maintenance Specifications
Procedure Number	18-018-016 (/qs3/pubsys2/xml/en/procedures/18/18-018-016.html)
Procedure Title	Fuel System
Update Reason	
Date	20-Jun-2007
Section	Section W - Warranty
Procedure Number	3381321 (/qs3/pubsys2/xml/en/warranty/3381321.html)
Procedure Title	All Engines United States and Canada Industrial (Off-Highway)
Update Reason	
Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	

Date	22-Oct-2007
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	99-001-052 (/qs3/pubsys2/xml/en/procedures/99/99-001-052.html)
Procedure Title	Vibration Damper, Viscous
Update Reason	
Date	27-Mar-2008
Section	Section W - Warranty
Procedure Number	3381310 (/qs3/pubsys2/xml/en/warranty/3381310.html)
Procedure Title	K19 Engines Australia and New Zealand Automotive
Update Reason	
Date	24-Apr-2008
Section	Section S - Service Assistance
Procedure Number	99-202-007 (/qs3/pubsys2/xml/en/procedures/99/99-202-007.html)
Procedure Title	Regional Offices - International
Update Reason	
Date	02-Dec-2008
Section	Section V - Maintenance Specifications
Procedure Number	18-018-017 (/qs3/pubsys2/xml/en/procedures/18/18-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	
Date	04-Dec-2008
Section	Section S - Service Assistance
Procedure Number	99-202-006 (/qs3/pubsys2/xml/en/procedures/99/99-202-006.html)
Procedure Title	Distributors and Branches
Update Reason	

Date	12-Dec-2008
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-010-037-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-010-037-om-ind.html)
Procedure Title	Turbocharger, Water Cooled
Update Reason	
Date	09-Jan-2009
Section	Section 1 - Operating Instructions
Procedure Number	00-101-018 (/qs3/pubsys2/xml/en/procedures/00/00-101-018.html)
Procedure Title	Starting Procedure After Extended Shutdown or Oil Change
Update Reason	
Date	25-Feb-2009
Section	Section S - Service Assistance
Procedure Number	99-202-003 (/qs3/pubsys2/xml/en/procedures/99/99-202-003.html)
Procedure Title	Emergency and Technical Service
Update Reason	
Date	01-Apr-2009
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	18-008-057-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-057-om-ind.html)
Procedure Title	Seawater Pump
Update Reason	

Date	12-May-2009
Section	Section W - Warranty
Procedure Number	3381775 (/qs3/pubsys2/xml/en/warranty/3381775.html)
Procedure Title	All Engines Worldwide Marine/Commercial Propulsion and Auxiliary Products
Update Reason	
Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-om (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-om.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	
Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Part Catalog
Update Reason	
Date	25-Aug-2009
Section	Section W - Warranty
Procedure Number	3381307 (/qs3/pubsys2/xml/en/warranty/3381307.html)
Procedure Title	All Diesel Engines Worldwide Generator Drive
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-204-002-om-indmar.html)
Procedure Title	About the Manual
Update Reason	

Date	16-Sep-2009
Section	Section 1 - Operating Instructions
Procedure Number	99-101-009-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-101-009-om-indmar.html)
Procedure Title	Engine Shutdown
Update Reason	
Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-204-007-om-indmar.html)
Procedure Title	General Repair Instructions
Update Reason	
Date	09-Oct-2009
Section	Section ES - Engine Storage
Procedure Number	99-000-005-om-ind (/qs3/pubsys2/xml/en/procedures/99/99-000-005-om-ind.html)
Procedure Title	Engine Storage - Long Term
Update Reason	
Date	13-Oct-2009
Section	Section 2 - Maintenance Guidelines
Procedure Number	99-102-001 (/qs3/pubsys2/xml/en/procedures/99/99-102-001.html)
Procedure Title	Maintenance Record Form
Update Reason	
Date	14-Oct-2009
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-007-043-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-007-043-om-indmar.html)
Procedure Title	Lubricating Oil Level
Update Reason	

Date	15-Oct-2009
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	99-008-045 (/qs3/pubsys2/xml/en/procedures/99/99-008-045.html)
Procedure Title	Radiator Hoses
Update Reason	
Date	03-Nov-2009
Section	Section 5 - Maintenance Procedures at 250 Hours or 6 Months
Procedure Number	99-008-046 (/qs3/pubsys2/xml/en/procedures/99/99-008-046.html)
Procedure Title	Supplemental Coolant Additive (SCA) and Antifreeze Concentration
Update Reason	
Date	04-Nov-2009
Section	Section 6 - Maintenance Procedures at 1500 Hours or 1 Year
Procedure Number	102-016-010-om (/qs3/pubsys2/xml/en/procedures/102/102-016-010-om.html)
Procedure Title	Engine Mounts
Update Reason	
Date	13-Nov-2009
Section	Section 1 - Operating Instructions
Procedure Number	00-101-004-om (/qs3/pubsys2/xml/en/procedures/00/00-101-004-om.html)
Procedure Title	Cold Weather Starting
Update Reason	

Date	17-Nov-2009
Section	Section 2 - Maintenance Guidelines
Procedure Number	99-102-999 (/qs3/pubsys2/xml/en/procedures/99/99-102-999.html)
Procedure Title	Maintenance Guidelines - Overview
Update Reason	
Date	18-Nov-2009
Section	Section 7 - Maintenance Procedures at 6000 Hours or 2 Years
Procedure Number	99-001-051 (/qs3/pubsys2/xml/en/procedures/99/99-001-051.html)
Procedure Title	Vibration Damper, Rubber
Update Reason	
Date	19-Nov-2009
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-008-066 (/qs3/pubsys2/xml/en/procedures/99/99-008-066.html)
Procedure Title	Coolant Level
Update Reason	
Date	19-Nov-2009
Section	Section S - Service Assistance
Procedure Number	99-202-002 (/qs3/pubsys2/xml/en/procedures/99/99-202-002.html)
Procedure Title	Routine Service and Parts
Update Reason	
Date	19-Nov-2009
Section	Section S - Service Assistance
Procedure Number	99-202-004 (/qs3/pubsys2/xml/en/procedures/99/99-202-004.html)
Procedure Title	Problem Solving
Update Reason	

Date	06-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-003-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-204-003-om-indmar.html)
Procedure Title	How to Use the Manual
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-204-006-om-indmar.html)
Procedure Title	General Safety Instructions
Update Reason	
Date	25-Jan-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-999-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-101-999-om-indmar.html)
Procedure Title	Operating Instructions - Overview
Update Reason	
Date	25-Jan-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-008-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-101-008-om-indmar.html)
Procedure Title	Engine Operating Range
Update Reason	
Date	25-Jan-2010
Section	Section 3 - Maintenance Procedures at Daily Interval
Procedure Number	99-103-002 (/qs3/pubsys2/xml/en/procedures/99/99-103-002.html)
Procedure Title	Drive Belts
Update Reason	

Date	25-Jan-2010
Section	Section V - Maintenance Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V - Maintenance Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	
Date	04-Feb-2010
Section	Section 1 - Operating Instructions
Procedure Number	99-101-015-om-indmar (/qs3/pubsys2/xml/en/procedures/99/99-101-015-om-indmar.html)
Procedure Title	Operating the Engine
Update Reason	
Date	18-Mar-2010
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-007-009 (/qs3/pubsys2/xml/en/procedures/18/18-007-009.html)
Procedure Title	Lubricating Oil Dipstick
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	

Date	07-Jan-2011
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-013-005-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-013-005-om-ind.html)
Procedure Title	Drive Belt, Alternator
Update Reason	Added information for captive fasteners on alternator belt guard.

Date	07-Jan-2011
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-008-001-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-001-om-ind.html)
Procedure Title	Belt Guard
Update Reason	Added to Section A.

Date	23-Mar-2011
Section	Section 2 - Maintenance Guidelines
Procedure Number	18-102-002-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-102-002-om-ind.html)
Procedure Title	Maintenance Schedule
Update Reason	Updated to todays standards with minor edits.

Date	28-Sep-2011
Section	Section A - Adjustment, Repair, and Replacement
Procedure Number	18-008-087-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-008-087-om-ind.html)
Procedure Title	Cooling Fan Belt Tensioner
Update Reason	Added procedure to outline and manual.

Date	20-Dec-2011
Section	Section V - Maintenance Specifications
Procedure Number	102-018-003 (/qs3/pubsys2/xml/en/procedures/102/102-018-003.html)
Procedure Title	Lubricating Oil Recommendations and Specifications
Update Reason	Edited to add the specific oils used in HHP Diesel engines, represented by CES (Cummins engineering standards) and API classification.
Date	25-May-2013
Section	Section 4 - Maintenance Procedures at Weekly Interval
Procedure Number	18-007-051 (/qs3/pubsys2/xml/en/procedures/18/18-007-051.html)
Procedure Title	Lubricating Oil Analysis
Update Reason	This procedure was added to the outline.
Date	28-Aug-2013
Section	Section V - Maintenance Specifications
Procedure Number	102-018-002 (/qs3/pubsys2/xml/en/procedures/102/102-018-002.html)
Procedure Title	Fuel Recommendations and Specifications
Update Reason	Updated fuel specifications.
Date	22-Oct-2013
Section	Section 2 - Maintenance Guidelines
Procedure Number	18-102-002-om-ind (/qs3/pubsys2/xml/en/procedures/18/18-102-002-om-ind.html)
Procedure Title	Maintenance Schedule
Update Reason	Changed typo on Maintenance Procedures at 5000 Hours or 2 Years to Maintenance Procedures at 6000 Hours or 2 Years.

Date	01-Apr-2015
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-016-002 (/qs3/pubsys2/xml/en/procedures/89/89-016-002.html)
Procedure Title	Engine Support Bracket, Front
Update Reason	Procedure has been updated with currently available grease recommendations.
Date	08-Apr-2015
Section	Section 8 - Cooling System - Group 08
Procedure Number	89-008-018-tr (/qs3/pubsys2/xml/en/procedures/89/89-008-018-tr.html)
Procedure Title	Cooling System
Update Reason	Added directions to allow the engine to cool before draining coolant.
Date	05-May-2015
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-033 (/qs3/pubsys2/xml/en/procedures/89/89-010-033.html)
Procedure Title	Turbocharger
Update Reason	Updated procedure to indicate that V-band clamps can not be reused.
Date	07-May-2015
Section	Section 0 - Product - Group 00
Procedure Number	99-000-005-tr (/qs3/pubsys2/xml/en/procedures/99/99-000-005-tr.html)
Procedure Title	Engine Storage - Long Term
Update Reason	Added to outline per corrective action.

Date	07-May-2015
Section	Section 0 - Product - Group 00
Procedure Number	99-000-009 (/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)
Procedure Title	Engine Steam Cleaning
Update Reason	Added to outline per corrective action.
Date	26-May-2015
Section	Section 5 - Fuel System - Group 05
Procedure Number	89-005-043 (/qs3/pubsys2/xml/en/procedures/89/89-005-043.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	Added this procedure to the manual.
Date	27-Oct-2015
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-014-008 (/qs3/pubsys2/xml/en/procedures/89/89-014-008.html)
Procedure Title	Engine Testing (In Chassis)
Update Reason	Procedure has been commonized for all high horsepower engine families.
Date	13-Jan-2016
Section	Section 1 - Cylinder Head - Group 01
Procedure Number	441-001-008 (/qs3/pubsys2/xml/en/procedures/441/441-001-008.html)
Procedure Title	Camshaft
Update Reason	Added runout measurements in the Inspect for Reuse section.

Date	24-Feb-2016
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-033 (/qs3/pubsys2/xml/en/procedures/89/89-010-033.html)
Procedure Title	Turbocharger
Update Reason	Updated the values for Compressor Impeller Radial End Clearance.

Date	11-Apr-2016
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-033 (/qs3/pubsys2/xml/en/procedures/89/89-010-033.html)
Procedure Title	Turbocharger
Update Reason	Added Measure, Disassembly, and Assembly steps to procedure.

Date	14-Jun-2016
Section	Section 1 - Cylinder Head - Group 01
Procedure Number	441-001-054 (/qs3/pubsys2/xml/en/procedures/441/441-001-054.html)
Procedure Title	Piston and Connecting Rod Assembly
Update Reason	Addition of punch marks to the procedure. The information was parented off of procedure 89-001-014.

Date	21-Jun-2016
Section	Section 3- Rocker Levers - Group 3
Procedure Number	89-003-009 (/qs3/pubsys2/xml/en/procedures/89/89-003-009.html)
Procedure Title	Rocker Lever Assembly
Update Reason	Updated rocker lever shaft outside diameter and valve rocker lever and injector rocker lever inside diameter specifications in the Inspect for Reuse section.

Date	02-Aug-2016
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-026 (/qs3/pubsys2/xml/en/procedures/89/89-001-026.html)
Procedure Title	Cylinder Block
Update Reason	Added oversized head gasket information to the Measure step.
Date	03-Nov-2016
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	Updated Lower Press Fit Area Outside Diameter MIN and MAX values for Standard and Oversize liners.
Date	04-Nov-2016
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-072-tr (/qs3/pubsys2/xml/en/procedures/89/89-010-072-tr.html)
Procedure Title	Air Intake Manifold Heater
Update Reason	Restructured entire procedure.
Date	22-Feb-2017
Section	Section 4 - Cam Followers/Tappets - Group 04
Procedure Number	89-004-001 (/qs3/pubsys2/xml/en/procedures/89/89-004-001.html)
Procedure Title	Cam Follower Assembly
Update Reason	Added clarification on the direction of the pressed-in ball for the cam follower shaft.

Date	17-Mar-2017
Section	Section V - Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	Added procedure to manual.
Date	17-Mar-2017
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	Added procedure to manual.
Date	24-Mar-2017
Section	Section 8 - Cooling System - Group 08
Procedure Number	89-008-002-tr (/qs3/pubsys2/xml/en/procedures/89/89-008-002-tr.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Updated to include an inspect and re-use guideline for the drive belt tensioner.
Date	22-Jun-2017
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-008 (/qs3/pubsys2/xml/en/procedures/89/89-006-008.html)
Procedure Title	Fuel Check Valve
Update Reason	New procedure added to this manual.
Date	05-Nov-2017
Section	Section 16 - Mounting Adaptations - Group 16
Procedure Number	89-016-005 (/qs3/pubsys2/xml/en/procedures/89/89-016-005.html)
Procedure Title	Flywheel
Update Reason	Added a Caution to the Install Step torque of the procedure.

Date	12-Dec-2017
Section	Section 8 - Cooling System
Procedure Number	89-008-002 (/qs3/pubsys2/xml/en/procedures/89/89-008-002.html)
Procedure Title	Drive Belt, Cooling Fan
Update Reason	Added Min/Max table for belt tension.
Date	03-Apr-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-036 (/qs3/pubsys2/xml/en/procedures/89/89-001-036.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	Added two caution statement. Added capscrew inspection criteria. Added to final step to punch mark the capscrews.
Date	13-Jul-2018
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-033-tr (/qs3/pubsys2/xml/en/procedures/89/89-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Corrected Schwitzer S500 Max inch conversion.
Date	20-Jul-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-047 (/qs3/pubsys2/xml/en/procedures/89/89-001-047.html)
Procedure Title	Piston Rings
Update Reason	Added Install Step.

Date	24-Jul-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	89-007-050 (/qs3/pubsys2/xml/en/procedures/89/89-007-050.html)
Procedure Title	Cylinder Smokes (Blue Smoke)
Update Reason	Added direction for the manifold to cool before removing.
Date	14-Nov-2018
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-008 (/qs3/pubsys2/xml/en/procedures/89/89-001-008.html)
Procedure Title	Camshaft
Update Reason	Updated the details for the removal and installation steps.
Date	21-Nov-2018
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	89-007-050 (/qs3/pubsys2/xml/en/procedures/89/89-007-050.html)
Procedure Title	Cylinder Smokes (Blue Smoke)
Update Reason	Procedure was updated to make it clear not to run the engine with the exhaust manual OFF.
Date	07-Jan-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-098 (/qs3/pubsys2/xml/en/procedures/89/89-001-098.html)
Procedure Title	Idler Gear Oil Pump and Water Pump
Update Reason	Updated the procedure to include the inspection process for the idler gear oil pump and water pump.

Date	07-Jan-2019
Section	Section 7 - Lubricating Oil System - Group 07
Procedure Number	89-007-031 (/qs3/pubsys2/xml/en/procedures/89/89-007-031.html)
Procedure Title	Lubricating Oil Pump
Update Reason	Updated the procedure includes the disassembly and clean for inspection process for the lubricating oil pump.

Date	18-Apr-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-023 (/qs3/pubsys2/xml/en/procedures/89/89-001-023.html)
Procedure Title	Crankshaft Seal, Front
Update Reason	Updated the procedure to include information regarding the installation of the crankshaft oil seal and dust seal using the front crankshaft seal service tool.

Date	18-Apr-2019
Section	Section 11 - Exhaust System - Group 11
Procedure Number	89-011-001 (/qs3/pubsys2/xml/en/procedures/89/89-011-001.html)
Procedure Title	Exhaust Connection Pipe
Update Reason	Updated the torque value in the Install step.

Date	26-Jun-2019
Section	Section 5 - Fuel System - Group 05
Procedure Number	89-005-016 (/qs3/pubsys2/xml/en/procedures/89/89-005-016.html)
Procedure Title	Fuel Pump
Update Reason	Updated the procedure to include the disassembly and assembly steps in order to rebuild the fuel pump.

Date	27-Jun-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-012 (/qs3/pubsys2/xml/en/procedures/89/89-001-012.html)
Procedure Title	Camshaft Gear (Camshaft Installed)
Update Reason	Updated the procedure to includes the correct service tool part numbers are included for the removal of the camshaft gear.

Date	27-Jun-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-013 (/qs3/pubsys2/xml/en/procedures/89/89-001-013.html)
Procedure Title	Camshaft Gear (Camshaft Removed)
Update Reason	Updated the procedure to includes the correct service tool part numbers are included for the removal of the camshaft gear.

Date	27-Jun-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-018 (/qs3/pubsys2/xml/en/procedures/89/89-001-018.html)
Procedure Title	Crankshaft Gear, Front (Crankshaft Installed)
Update Reason	Updated the procedure to include the inspect for reuse process and correct specifications.

Date	17-Jul-2019
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-020 (/qs3/pubsys2/xml/en/procedures/89/89-001-020.html)
Procedure Title	Crankshaft Gear, Rear (Crankshaft Installed)
Update Reason	Updated the procedure includes the correct service tool part numbers, instructions and inspect for reuse.

Date	05-Aug-2019
Section	Section 1 - Cylinder Block
Procedure Number	89-001-031 (/qs3/pubsys2/xml/en/procedures/89/89-001-031.html)
Procedure Title	Gear Cover, Front
Update Reason	Added statement to Clean step: Inspect and clean off the gasket sealant from tapped blind holes, if present.

Date	24-Sep-2019
Section	Section 4 - Cam Followers/Tappets
Procedure Number	89-004-001 (/qs3/pubsys2/xml/en/procedures/89/89-004-001.html)
Procedure Title	Cam Follower Assembly
Update Reason	Added Lever Bore Inside Diameter specifications table.

Date	23-Oct-2019
Section	Section 7 - Lubricating Oil System
Procedure Number	89-007-007 (/qs3/pubsys2/xml/en/procedures/89/89-007-007.html)
Procedure Title	Lubricating Oil Cooler Element
Update Reason	Updated torque sequence for the oil cooler bolts.

Date	23-Oct-2019
Section	Section 1 - Cylinder Block
Procedure Number	89-001-036 (/qs3/pubsys2/xml/en/procedures/89/89-001-036.html)
Procedure Title	Idler Gear, Camshaft
Update Reason	Added Clean and Inspect for Reuse step.

Date	17-Jan-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-028 (/qs3/pubsys2/xml/en/procedures/89/89-001-028.html)
Procedure Title	Cylinder Liner
Update Reason	Removed inaccurate statement from the Install steps of the procedure. The statement noted that the inside diameter could not surpass 0.076 mm [0.003 in] out-of-round at the top.
Date	03-Feb-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-022-001_14 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_14.html)
Procedure Title	Service Tools
Update Reason	Added cylinder diagnostic tool to the Service Tools table.
Date	25-Mar-2020
Section	Section 14 - Engine Testing - Group 14
Procedure Number	00-014-012 (/qs3/pubsys2/xml/en/procedures/00/00-014-012.html)
Procedure Title	Engine Compression
Update Reason	Added additional information to the procedure.
Date	10-Aug-2020
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-003 (/qs3/pubsys2/xml/en/procedures/89/89-013-003.html)
Procedure Title	Charging System Alternator Bracket
Update Reason	Added procedure to manual.

Date	13-Aug-2020
Section	Section 13 - Electrical Equipment - Group 13
Procedure Number	89-013-005 (/qs3/pubsys2/xml/en/procedures/89/89-013-005.html)
Procedure Title	Charging System Alternator Drive Belt
Update Reason	Added step for adjusting the v-ribbed belt on Engines with a 140A alternator.

Date	13-Aug-2020
Section	Section 9 - Drive Units - Group 09
Procedure Number	89-009-010 (/qs3/pubsys2/xml/en/procedures/89/89-009-010.html)
Procedure Title	Charging System Alternator Drive Pulley
Update Reason	Added step for installing the alternator drive pulley on engines with a 75A alternator.

Date	13-Aug-2020
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-022-001_01 (/qs3/pubsys2/xml/en/procedures/89/89-022-001_01.html)
Procedure Title	Service Tools
Update Reason	Added Part Number 3164067, Loctite® 5699 RTV Sealant to Select Service Tools list.

Date	13-Aug-2020
Section	Section V - Specifications
Procedure Number	89-018-017 (/qs3/pubsys2/xml/en/procedures/89/89-018-017.html)
Procedure Title	Lubricating Oil System
Update Reason	Revised table to include accurate information.

Date	09-Jul-2021
Section	Section 2 - Cylinder Head - Group 02
Procedure Number	89-002-004-tr (/qs3/pubsys2/xml/en/procedures/89/89-002-004-tr.html)
Procedure Title	Cylinder Head
Update Reason	Added note indicating the valve head must not be more than 0.39 mm [0.015 in] below the surface of the cylinder head or 0.21 mm [0.008 in] above the surface of the cylinder head.

Date	14-Jul-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-023 (/qs3/pubsys2/xml/en/procedures/89/89-001-023.html)
Procedure Title	Crankshaft Seal, Front
Update Reason	Added Lubriplate™ 105, Part Number 3163086, or equivalent to Select Service Tools list. Added instruction to apply Lubriplate™ 105, Part Number 3163086, or equivalent, onto the remover screw threads and tip.

Date	11-Oct-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-020 (/qs3/pubsys2/xml/en/procedures/89/89-001-020.html)
Procedure Title	Crankshaft Gear, Rear (Crankshaft Installed)
Update Reason	Added Select Service Tools list to procedure. Revised and corrected Remove steps to include only necessary information. Corrected minimum and maximum specifications table for Crankshaft Outside Diameter (Gear Location).

Date	01-Dec-2021
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-024 (/qs3/pubsys2/xml/en/procedures/89/89-001-024.html)
Procedure Title	Crankshaft Seal, Rear
Update Reason	Added mention of Lubriplate™ 105, Part Number 3163086, or equivalent throughout procedure. Revised Measure and Remove steps.
Date	12-Apr-2022
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-098 (/qs3/pubsys2/xml/en/procedures/89/89-001-098.html)
Procedure Title	Idle Gear Oil Pump and Water Pump
Update Reason	Minor wording correction made to procedure. Added gear brushing to step that calls for measuring the inside and outside diameter of the idler.
Date	18-Aug-2022
Section	Section 5 - Fuel System - Group 05
Procedure Number	89-005-043 (/qs3/pubsys2/xml/en/procedures/89/89-005-043.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	Added Voltage Check step to procedure.
Date	28-Sep-2022
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	89-003-013 (/qs3/pubsys2/xml/en/procedures/89/89-003-013.html)
Procedure Title	Rocker Lever Housing
Update Reason	Added instruction to inspect the threaded holes for damage in the Clean and Inspect for Reuse steps. Added instruction to install the o-ring on the groove of the rocker lever housing in the Install steps.

Date	27-Jan-2023
Section	Section 14 - Engine Testing - Group 14
Procedure Number	89-014-005 (/qs3/pubsys2/xml/en/procedures/89/89-014-005.html)
Procedure Title	Engine Testing (Engine Dynamometer)
Update Reason	Revised blowby specification tables to remove reference to CPL and rpm.

Date	27-Jan-2023
Section	Section 6 - Injectors and Fuel Lines - Group 06
Procedure Number	89-006-003 (/qs3/pubsys2/xml/en/procedures/89/89-006-003.html)
Procedure Title	Air in Fuel
Update Reason	Deleted and added text to revise actions to take if an air leak is found.

Date	08-Feb-2023
Section	Section 1 - Cylinder Block - Group 01
Procedure Number	89-001-036 (/qs3/pubsys2/xml/en/procedures/89/89-001-036.html)
Procedure Title	Idle Gear, Camshaft
Update Reason	Corrected idler shaft outer diameter minimum/maximum specification table.

Date	13-Sep-2023
Section	Section 3 - Rocker Levers - Group 03
Procedure Number	89-003-006 (/qs3/pubsys2/xml/en/procedures/89/89-003-006.html)
Procedure Title	Overhead Set (OBC)
Update Reason	Table misaligned in procedure, edited procedure to correct table.

Date	25-Oct-2023
Section	Section 10 - Air Intake System - Group 10
Procedure Number	89-010-033-tr (/qs3/pubsys2/xml/en/procedures/89/89-010-033-tr.html)
Procedure Title	Turbocharger
Update Reason	Corrected torque value listed in the second Assemble stepblock.

Last Modified: 09-Nov-2023
